

MAT111 Assignment

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Problem 1

From the definition, $xRy \iff x = y$ ($\star$).

Reflexive: xRx by definition. Thus, R is reflexive.

Symmetric: By ($\star$), xRy is equivalent to xRx . Since R is reflexive, it is also symmetric.

Transitive: If xRy and yRz , $x = y = z$ by applying ($\star$) twice, so xRz . Thus, R is transitive.

R is therefore an equivalence relation.

Problem 2

From the definition, $xRy \iff x = n + k$ and $y = m + k$ where $n, m \in \mathbb{Z}, k \in \mathbb{R}$ ($\star$).

Reflexive: xRx since $x - x = 0 \in \mathbb{Z}$, so R is reflexive.

Symmetric: If xRy , $x - y = n - m \in \mathbb{Z}$ by ($\star$). $yRx \iff y - x \in \mathbb{Z}$ but $y - x = m - n \in \mathbb{Z}$ so R is symmetric.

Transitive: If xRy and yRz , $x - y \in \mathbb{Z}$ and $y - z \in \mathbb{Z}$. Adding these 2 statements gives $x - y + y - z$. Since $\mathbb{Z}$ is closed under addition, $x - z \in \mathbb{Z}$ so R is transitive.

R is therefore an equivalence relation.

Problem 3

Let such a relation be R . If xRy then yRx since R is symmetric. If xRy and yRx , then xRx since R is transitive. Thus R must be reflexive.

Problem 4

Let R be the set of all possible relations on A .

$$R = \{S : S \subseteq A \times A\}$$

1	$\{\}$
2	$\{(a, a)\}$
3	$\{(a, b)\}$
4	$\{(b, a)\}$
5	$\{(b, b)\}$
6	$\{(a, a), (a, b)\}$
7	$\{(a, a), (b, a)\}$
8	$\{(a, a), (b, b)\}$
9	$\{(a, b), (b, a)\}$
10	$\{(a, b), (b, b)\}$
11	$\{(b, a), (b, b)\}$
12	$\{(a, a), (a, b), (b, a)\}$
13	$\{(a, a), (a, b), (b, b)\}$
14	$\{(a, a), (b, a), (b, b)\}$
15	$\{(a, b), (b, a), (b, b)\}$
16	$\{(a, a), (a, b), (b, a), (b, b)\}$

Problem 5

Reflexive: $|x - x| = 0 \leq 1$ so R is reflexive.

Symmetric: $|x - y| = |y - x|$ so R is symmetric.

Transitive: By counter-example of $x = 1, y = 2, z = 3$, R is not transitive.

Problem 6

x	y	$x^2 + y^2$
even	even	even
even	odd	odd
odd	even	odd
odd	odd	even

Therefore, equivalently $xRy \iff x \equiv y \pmod{2}$.

Reflexive: $x \equiv x \pmod{2}$ so R is reflexive.

Symmetric: $x \equiv y \pmod{2}$ is equivalent to $y \equiv x \pmod{2}$ so R is symmetric.

Transitive: If $x \equiv y \pmod{2}$ and $y \equiv z \pmod{2}$, adding both equations gives $x + y \equiv y + z \pmod{2}$ and $x \equiv z \pmod{2}$ follows. Thus, R is transitive.

Since R is reflexive, symmetric and transitive, it is an equivalence relation.

Problem 7

$$|R| = \text{number of relations in an equivalence class} \times \text{number of equivalence classes}$$

All m elements of an equivalence class are related to all m elements gives m^2 relations in an

equivalence class.

All equivalence classes are disjoint so there are $|A|/m$ equivalence classes.

$$|R| = m^2 \cdot \frac{|A|}{m} = m|A|$$

Problem 8

$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \equiv y \pmod{2}\}$ has 2 (finite number) equivalence classes.

Problem 9

$$4^{|A|}$$

Problem 10

Let $S_i = \{4n + i : n \in \mathbb{Z}\}$.

$$\mathbb{Z} = \bigcup_{i=0}^4 S_i$$

Problem 11

$$R = \{(x, y) : x, y \in \mathbb{Z}, x \equiv y \pmod{2}\}$$

Problem 12

Addition:

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Multiplication:

$\times$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	4

Problem 13

If $a \equiv b \pmod{n}$ then $a = pn + i$ and $b = qn + i$ where $p, q, i \in \mathbb{Z}$ and $0 \leq i < n$.

If $c \equiv d \pmod{n}$ then $c = un + j$ and $d = vn + j$ where $u, v, j \in \mathbb{Z}$ and $0 \leq j < n$.

Therefore, $a + c \equiv i + j \pmod{n}$ and $i + j \equiv b + d \pmod{n}$. Since $\pmod{n}$ satisfies transitivity, we're done.

Problem 14

If $a \equiv b \pmod{n}$ then $a = pn + i$ and $b = qn + i$ where $p, q, i \in \mathbb{Z}$ and $0 \leq i < n$.

If $c \equiv d \pmod{n}$ then $c = un + j$ and $d = vn + j$ where $u, v, j \in \mathbb{Z}$ and $0 \leq j < n$.

Therefore, $a \cdot c \equiv i \cdot j \pmod{n}$ and $i \cdot j \equiv b \cdot d \pmod{n}$. Since $\pmod{n}$ satisfies transitivity, we're done.

Problem 15

$ab \not\equiv 0 \pmod{p} \forall a, b \in [1, p-1]$ by the definition of a prime number.

Lemma 0 $ab \not\equiv ac \pmod{p}$ for $a, b, c \in [1, p-1], b \neq c$.

Proof. Assume $ab \equiv ac \pmod{p}$ for $b \neq c$. Then $a(b - c) \equiv 0 \pmod{p}$ which was shown to be false earlier.

Let $S_a = \{ab \pmod{p} : b \in [1, p-1]\}$ for any $p \nmid a$. Then $|S_a| = p-1$ by **Lemma 0**. Since $0 \notin S_a$, it follows that $S_a = \mathbb{Z}_p \setminus \{0\} \ni 1$.

Problem 16

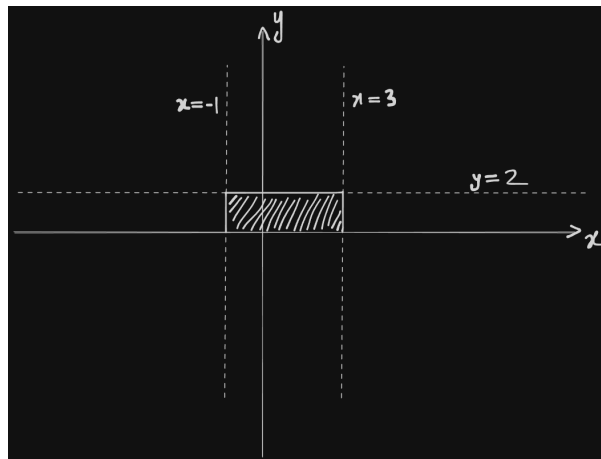


Figure 1: $X \in [-1, 3] \times [0, 2]$

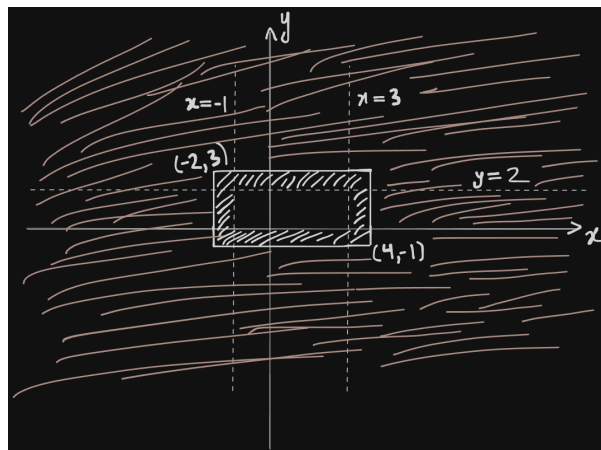


Figure 2: $X^{\mathbb{C}}$ and $X^{\mathbb{C}} \cap [-2, 4] \times [-1, 3]$

Problem 17

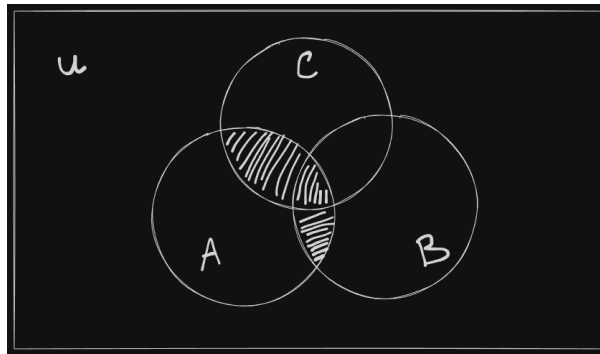


Figure 3: $(A \cap B) \cup (A \cap C)$

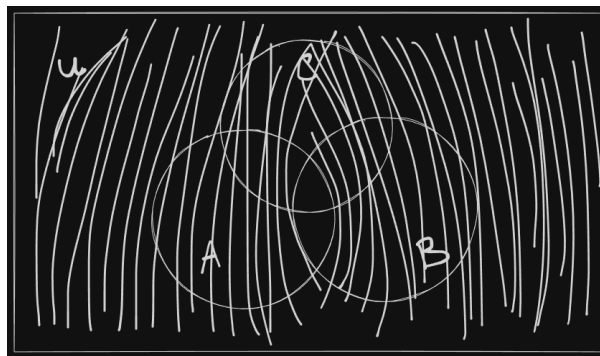


Figure 4: $\overline{A \cap B}$

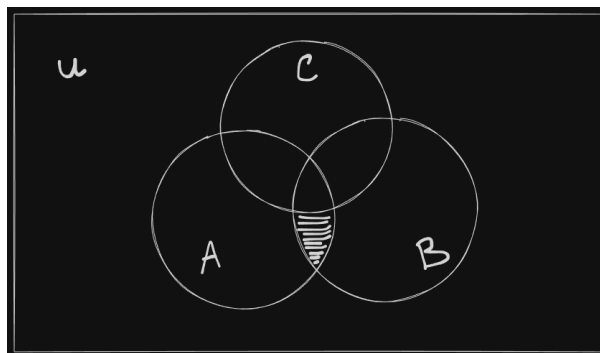


Figure 5: $(A \cap B) - C$

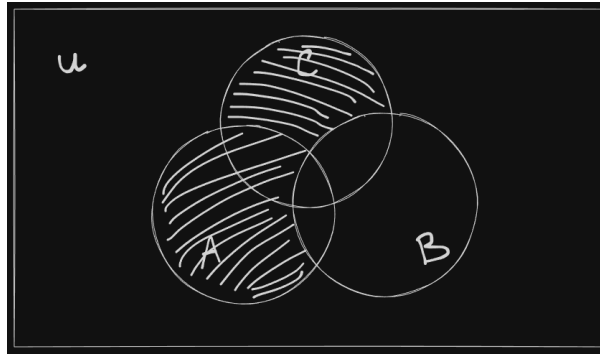


Figure 6: $(A - B) \cup C$

Problem 18

First statement is true.

Second statement is false.

Third statement is true.

Problem 19

Only $\mathbb{N}$ has a smallest element. None have a greatest element. Only $\mathbb{N}$ has the well-ordering property.

Problem 20

$$S = \{x : x \in \mathbb{R}, x\}$$

Problem 21

$$\sqrt{2}$$

Problem 22

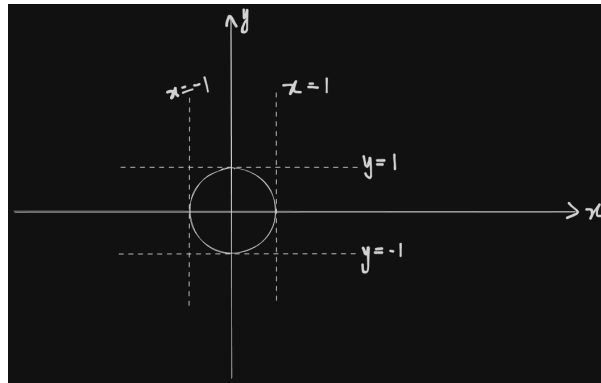


Figure 7: $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\}$

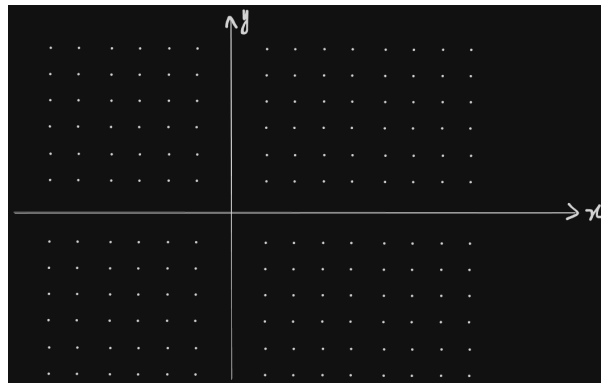


Figure 8: $\mathbb{Z} \times \mathbb{Z}$

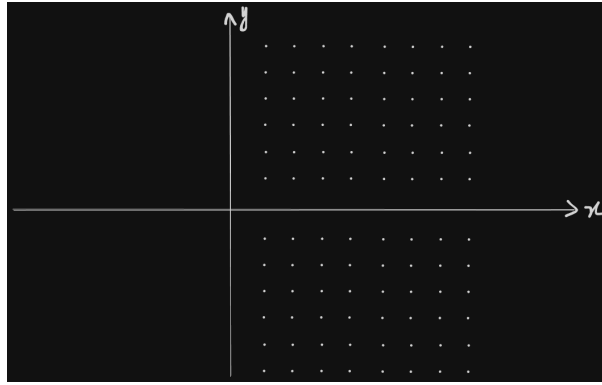


Figure 9: $\mathbb{N} \times \mathbb{Z}$

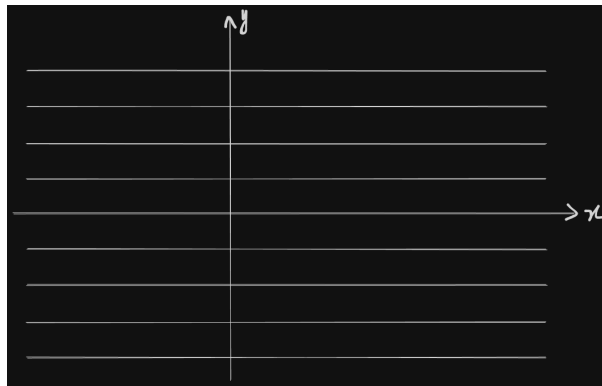


Figure 10: $\mathbb{R} \times \mathbb{Z}$

Problem 23

Because $\mathbb{N} \subseteq \mathbb{R}$.

Problem 24

$$|\{X : X \in \mathcal{P}(\{1, 2, 3\}), 2 \in X\}| = 2^2 = 4$$

$$|\{X : X \in \mathcal{P}(A), |X| \leq 1\}| = m + 1$$

$$|\{X : X \subseteq \mathcal{P}(A), |X| \leq 1\}| = 1 + 2^m$$

Problem 25

Yes. No.

Problem 27

First statement: For all integers n , there exists an integer $m = n + 5$ is true because there are infinitely many integers.

Second statement: There exists an integer $m = n + 5$ for all integers n is true because m cannot be $1 + 5$ and $2 + 5$ at the same time.